

EcoSphere Database Blueprint (MongoDB)

Design Principles

Single User collection with role-based access.

Feature-based collections.

Cloudinary stores files; MongoDB stores URLs only.

Reports are generated dynamically.

users

Field	Type	Purpose
_id	ObjectId	PK
name	String	Full name
email	String	Unique
password	String	Hash
role	Enum	admin/manager/employee/auditor
department	ObjectId	Department
designation	String	Role title
phone	String	Optional
profileImage	Object	Cloudinary
xp	Number	XP
coins	Number	Coins
badges	ObjectId[]	Badge refs
acceptedPolicies	ObjectId[]	Accepted
status	Enum	active/inactive
createdAt	Date	
updatedAt	Date	

departments

Field	Type	Purpose
_id	ObjectId	PK
name	String	Department
manager	ObjectId	User ref
description	String	
employeeCount	Number	Cache
environmentScore	Number	
socialScore	Number	
governanceScore	Number	
overallScore	Number	
status	Enum	

operations

Field	Type	Purpose
_id	ObjectId	PK
department	ObjectId	Dept
manager	ObjectId	Creator
type	Enum	Purchase etc
category	String	Diesel etc
quantity	Number	
unit	String	
cost	Number	
proof	Object	Cloudinary
remarks	String	
carbonEmission	Number	Calculated
date	Date	

emissionFactors

Field	Type	Purpose
_id	ObjectId	PK
name	String	
category	String	
factor	Number	
unit	String	
description	String	

policies

Field	Type	Purpose
_id	ObjectId	PK
title	String	
description	String	
department	ObjectId	
attachment	Object	PDF
createdBy	ObjectId	Admin
effectiveDate	Date	
status	Enum	

csrActivities

Field	Type	Purpose
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_id	ObjectId	PK
title	String	
description	String	
department	ObjectId	
createdBy	ObjectId	
rewardXP	Number	
rewardCoins	Number	
location	String	
date	Date	
proofRequired	Boolean	
participants	Array	Embedded
status	Enum	

challenges

Field	Type	Purpose
_id	ObjectId	PK
title	String	
description	String	
department	ObjectId	
difficulty	Enum	
rewardXP	Number	
rewardCoins	Number	
startDate	Date	
endDate	Date	
proofRequired	Boolean	
participants	Array	Embedded
status	Enum	

rewards

Field	Type	Purpose
_id	ObjectId	PK
title	String	
description	String	
coinCost	Number	
stock	Number	
image	Object	
status	Enum	

badges

Field	Type	Purpose
_id	ObjectId	PK
title	String	
description	String	
icon	Object	
rule	String	Unlock rule
status	Enum	

audits

Field	Type	Purpose
_id	ObjectId	PK
department	ObjectId	
auditor	ObjectId	
date	Date	
remarks	String	
overallStatus	Enum	

compliancelssues

Field	Type	Purpose
_id	ObjectId	PK
audit	ObjectId	
department	ObjectId	
owner	ObjectId	
title	String	
description	String	
severity	Enum	
deadline	Date	
status	Enum	
proof	Object	
resolvedDate	Date	

notifications

Field	Type	Purpose
_id	ObjectId	PK
recipient	ObjectId	
title	String	

message	String	
type	Enum	
link	String	
isRead	Boolean	
createdAt	Date	

Recommended Indexes

users.email (unique), users.role, departments.name (unique), operations(department,date), notifications(recipient,isRead), complianceIssues(status,deadline).

Embedding Strategy

Embed CSR participants and challenge participants. Use references for users, departments, audits and policies. Generate reports and leaderboards dynamically.